Does the IVR Signal Survive Full Momentum Reconstruction?

A detector-level y-pol closure study with ^{112}Sn and ^{124}Sn
190 MeV/u with SAMURAI + NEBULA Plus

Baiting Tian

Tsinghua University & RIKEN

SAMURAI International Collaboration Workshop 2026

What Are We Trying to Establish?

Physics question

Can the isovector reorientation (IVR) signal from a y-polarized deuteron remain observable after the realistic SAMURAI detector and momentum-reconstruction chain?

Why two isotopes?

^{112}Sn provides an isotope reference.
^{124}Sn provides the stronger neutron-rich response.

Current answer

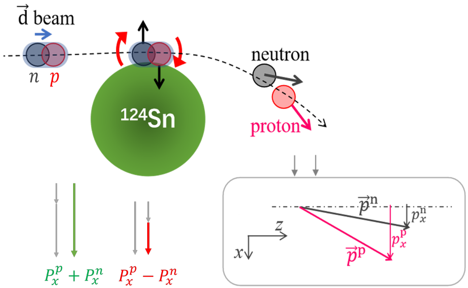
For ^{124}Sn, the γ-dependent ordering survives reconstructed proton, neutron, and event-plane determination in a controlled neutron fiducial region.

Scope

The observable is fully reconstructed, while the present event-quality selection is still truth-assisted.

This is a detector-level closure study, not yet a complete experimental uncertainty analysis.

Physical Picture: Isovector Reorientation



- A polarized deuteron approaches a neutron-rich target.
- Proton and neutron experience different isovector mean-field forces.
- The breakup fragments acquire a polarization-dependent relative momentum bias.

$$E_{\mathrm{sym}}(\rho) = E_{\mathrm{sym}}^{\mathrm{kin}}(\rho) + C\left(\frac{\rho}{\rho_0}\right)^{\gamma}$$

Model parameter

γ parametrizes the density dependence used in the ImQMD calculation.

Why Compare Two Tin Isotopes?

target	Z	N	role
^{112}Sn	50	62	isotope reference
^{124}Sn	50	74	primary sensitivity

- Same proton number controls much of the target charge and kinematic environment.
- Different neutron excess changes the isovector mean-field response.
- The same detector/reconstruction chain is required for a controlled isotope comparison.

Takeaway

A common detector effect should influence both isotopes, while an isovector response should change with target neutron excess. ^{124}Sn is the primary sensitivity target; ^{112}Sn is the isotope reference.

Why Start with y Polarization?

- y-pol and z-pol are both physically valuable IVR channels.
- y-pol currently has the most mature combined simulation, detector folding, momentum reconstruction, and analysis baseline.
- y-pol also has an established left-right-up-down (LRUD) beam-monitoring strategy for vertical tensor polarization p_{yy}.

First-stage baseline

y-pol provides a realistic first-stage path to testing the IVR concept at SAMURAI.

z-pol

z-pol remains a second-stage extension after successful commissioning.

Reconstructed IVR Observable

$$\phi_{\mathrm{event}}^{\mathrm{reco}} = \mathrm{atan2}\left(p_{y,p}^{\mathrm{reco}} + p_{y,n}^{\mathrm{reco}}, p_{x,p}^{\mathrm{reco}} + p_{x,n}^{\mathrm{reco}}\right)$$

$$\Delta p_{x,\mathrm{reco}}^{\mathrm{rot}} = p_{x,p,\mathrm{reco}}^{\mathrm{rot}} - p_{x,n,\mathrm{reco}}^{\mathrm{rot}}$$

$$R_x^{\mathrm{reco,fid}} = \frac{N(\Delta p_{x,\mathrm{reco}}^{\mathrm{rot}} > 0)}{N(\Delta p_{x,\mathrm{reco}}^{\mathrm{rot}} < 0)}$$

Experimental comparison

$$R_{x,\mathrm{data}}^{\mathrm{reco,fid}} \quad \text{vs.} \quad R_{x,\mathrm{model}}^{\mathrm{fold,fid}}(\gamma)$$

This is a reconstructed fiducial observable, not a global truth-level ratio.

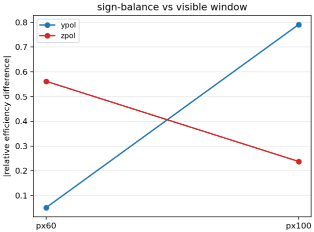
From ImQMD Events to the Measured Observable

ImQMD $^{112}\mathrm{Sn}/^{124}\mathrm{Sn}$ $\gamma = 0.5 \ldots 0.8$ → Geant4 detector response → PDC proton reco → NEBULA + Plus neutron reco → reco event plane → reco px60 $R_x^{\mathrm{reco,fid}}$

$$N_{\mathrm{model,reco},j}(\gamma) = \sum_i K_{ji} N_{\mathrm{model,true},i}(\gamma) + b_j$$

Current closure	Final experimental analysis
reco observable + truth-assisted quality cut	reco observable + reco-only quality cut

Defining a Controlled Neutron Fiducial Region



Neutron acceptance is strongly phase-space dependent; full phase space contains near-zero-efficiency regions.

$$\epsilon_+ = 0.678, \qquad \epsilon_- = 0.644$$

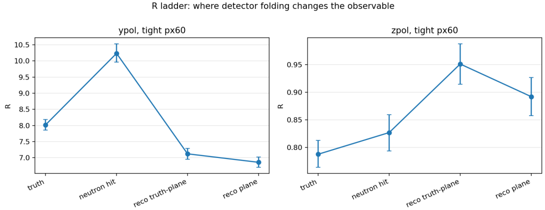
These are neutron-hit efficiencies in the truth-side $|p_{x,n}| < 60$ MeV$/c$ fiducial, pooled over y-pol tight events and all γ samples.

px60

px60 means the reconstructed neutron fiducial $|p_{x,n}^{\mathrm{reco}}| < 60$ MeV$/c$.

Takeaway: px60 defines a fiducial detector-level measurement. It does not reconstruct the unobserved full phase space.

Where Does Detector Folding Change the Observable?

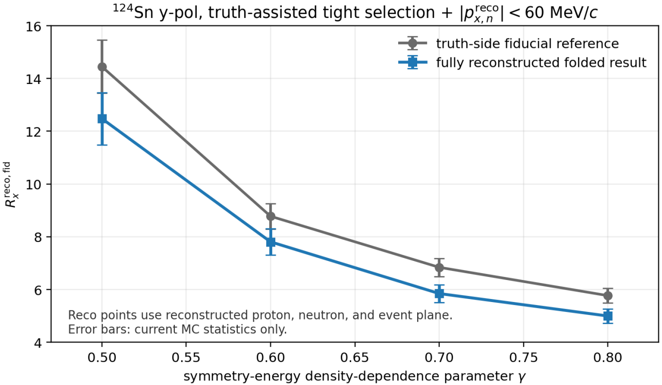


1. Neutron visibility/acceptance produces the largest shift.
2. Proton momentum reconstruction is not the dominant bottleneck.
3. Event-plane reconstruction adds a smaller correction.

Takeaway

The absolute ratio changes, but the γ ordering remains.

Primary Sensitivity: ^{124}Sn



Current closure selection: truth-assisted tight event-quality cut + reconstructed px60 fiducial.

Takeaway

For ^{124}Sn, the monotonic γ ordering survives the complete proton, neutron, and event-plane reconstruction chain. Error bars are current MC statistics only.

Isotope Reference: ^{112}Sn versus ^{124}Sn

^{124}Sn

Primary neutron-rich sensitivity target. Same-chain y-pol tight px60 full-reco closure is available and shown in the main result.

Current audit

The production description includes Sn112 + Sn124, but the locally exported tight-px60 folded tables are Sn124-only.

^{112}Sn

Isotope reference target. It tests whether the response follows target neutron excess rather than only generic kinematics or detector effects.

Consequence

A direct quantitative isotope plot is not shown in the main result yet. Full folded Sn112 closure with the identical chain remains to be completed.

A common detector effect should influence both isotopes; an isovector response should change with target neutron excess.

Statistical Scale versus Experimental Systematics

Closest adjacent interval in the current ^{124}Sn y-pol folded result:

$$\gamma = 0.7 \to 0.8, \qquad N_{\mathrm{stat},3\sigma} \simeq 2.8 \times 10^3.$$

Planning scale

Current planning anchor gives about 2.75×10^5 usable y-pol events for ^{124}Sn.

Not included

Detector, polarization, background, and model uncertainties are not included in this statistical-only estimate.

Systematic priorities

neutron efficiency; threshold and time-of-flight T_0; cross-talk and fake neutrons; backgrounds; p_{yy} polarization; detector alignment; model dependence; reco-only selection migration.

Current Status and Next Closure Step

Established

- y-pol IVR observable is reconstructable.
- ^{124}Sn γ ordering survives detector folding.
- px60 provides a controlled fiducial region.
- folded model comparison is the right strategy.

Still required

- reco-only event-quality cuts,
- full Sn112/Sn124 common-chain closure,
- selection efficiency, purity, and migration,
- polarization and detector-systematic pseudo-data tests.

The next decisive analysis is not another truth-level calculation. It is a fully reco-only isotope closure test.

Summary

1. IVR produces a polarization-dependent proton-neutron relative-momentum bias near a neutron-rich target.
2. ^{124}Sn is the primary sensitivity target, while ^{112}Sn provides a controlled isotope reference.
3. The γ-dependent ordering survives realistic detector folding and full proton, neutron, and event-plane reconstruction in the px60 fiducial region for ^{124}Sn.
4. The experiment should compare reconstructed data with identically folded model predictions; the next step is reco-only isotope closure and detector-systematic validation.

Bottom line

The present study establishes detector-level observability, not yet a complete experimental uncertainty budget. A y-pol measurement provides a realistic first-stage path to testing the IVR concept at SAMURAI.

Thanks

Questions?

Backup Roadmap

- A: model and observable
- B: reconstruction
- C: neutron acceptance
- D: ^{124}Sn full numbers
- E: ^{112}Sn and isotope comparison
- F: z-pol extension
- G: polarization monitoring
- H: systematics and future closure

Backup A: Observable and Bounded Asymmetry

$$\Delta p^{\mathrm{rot}}_{x,\mathrm{reco}} = p^{\mathrm{rot}}_{x,p,\mathrm{reco}} - p^{\mathrm{rot}}_{x,n,\mathrm{reco}}, \qquad R^{\mathrm{reco,fid}}_x = \frac{N(\Delta p^{\mathrm{rot}}_{x,\mathrm{reco}} > 0)}{N(\Delta p^{\mathrm{rot}}_{x,\mathrm{reco}} < 0)}.$$

$$A_x = \frac{N_+ - N_-}{N_+ + N_-} = \frac{R_x - 1}{R_x + 1}.$$

Vocabulary

Truth, reconstructed, folded, and unfolded observables are different objects. The main result uses a reconstructed fiducial observable.

Backup A: Folding Vocabulary

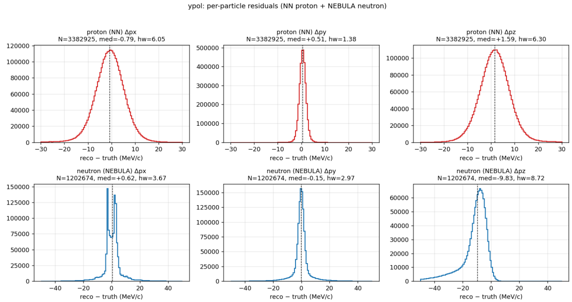
$$N_{\mathrm{reco},j} = \sum_i K_{ji} N_{\mathrm{true},i} + b_j .$$

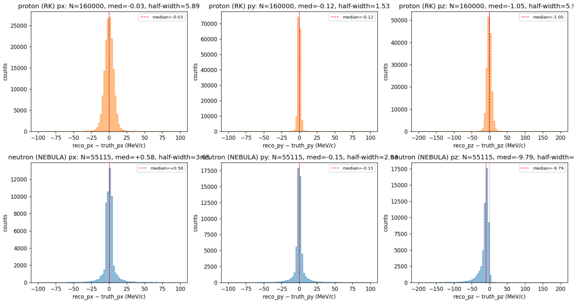
- truth: generator-level kinematics after physics selection,
- reconstructed: variables built from detector hits,
- folded: model events passed through detector response and reconstruction,
- unfolded: estimator of truth after response inversion and regularization.

Current limitation

The event-quality cut is still truth-assisted; the neutron px60 fiducial and the observable numerator/denominator are reconstructed.

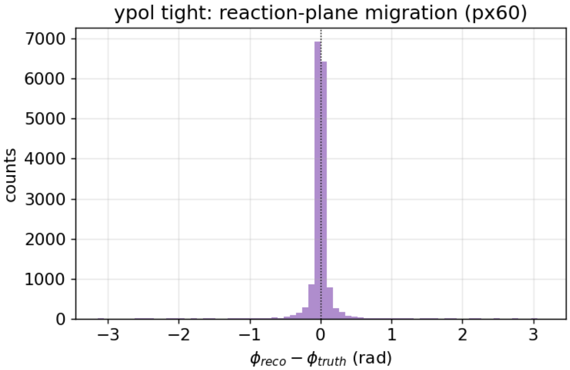
Backup B: Proton and Neutron Residuals

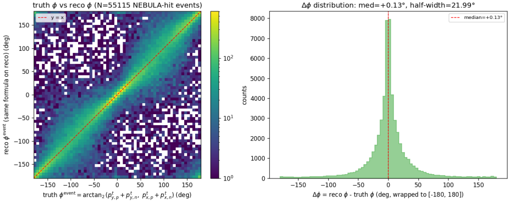




Left: NN breakup diagnostic. Right: earlier RK+NEBULA y-pol full-reco study. Half-width and RMSE should not be mixed; tails matter for RMSE.

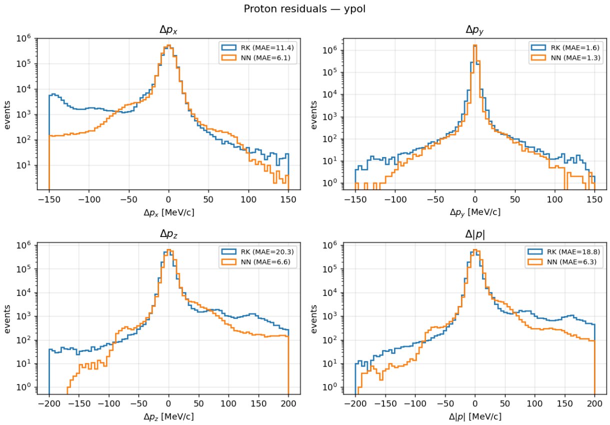
Backup B: Event-Plane Resolution





Reaction-plane migration broadens the observable but is not the leading change in the R ladder.

Backup B: RK vs NN Proton Cross-Check



Strict RK–NN comparison is available on the common configuration without NEBULA Plus.

y-pol metric	RK	NN
Δp_x half-width	5.86	6.01
Δp_x RMSE	39.7	13.3
Δp_z half-width	5.88	6.25
Δp_z RMSE	87.8	15.7

Proton reconstruction is cross-checked and is not the dominant observable shift.

Backup B: NN Proton Backend

Inputs

PDC1/PDC2 hit positions:

$$(x_1, y_1, z_1, x_2, y_2, z_2).$$

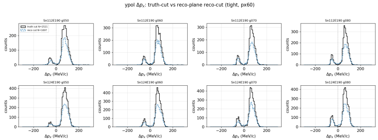
Outputs

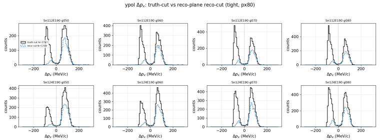
Target-frame proton momentum:

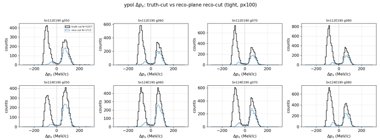
$$(p_x, p_y, p_z)_{\mathrm{target}}.$$

- MLP backend used through the canonical reconstruction CLI.
- Training domain covers the breakup kinematics used here.
- NN has comparable core resolution to RK and smaller catastrophic tails in the common-geometry comparison.

Backup C: px60, px80, px100



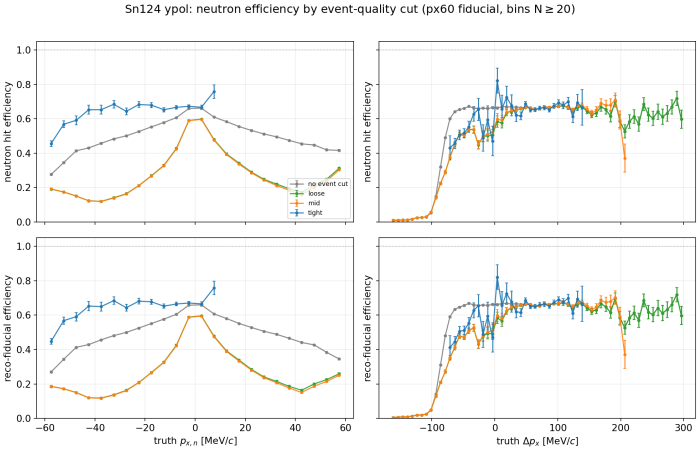




px60 px80 px100

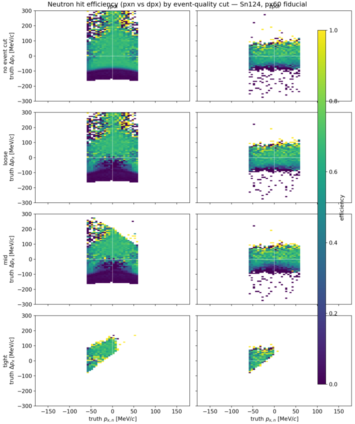
px80 improves over px100, but px60 removes most of the low-efficiency negative-$p_{x,n}$ branch.

Backup C: 1D Neutron Efficiency



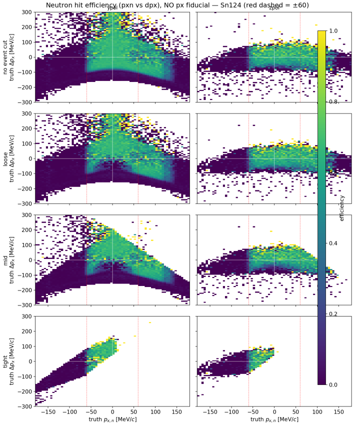
Efficiency shape is similar across event-quality cuts; cuts mainly change which phase-space region is populated.

Backup C: 2D Neutron Efficiency



The response is not uniform in $(p_{x,n}, \Delta p_x)$; this motivates folded model comparison.

Backup C: No-Fiducial Efficiency Landscape

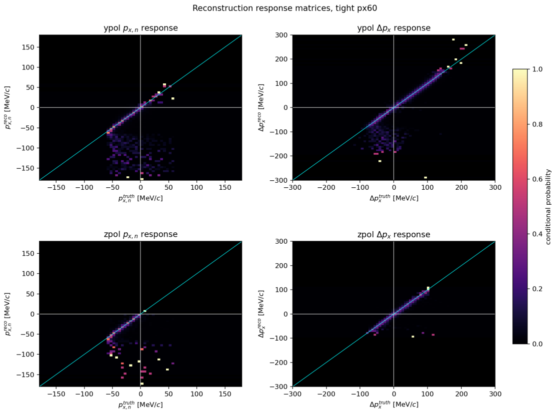


Near-zero-efficiency regions make a simple $1/\epsilon$ correction or raw unfolding unstable.

Backup C: Sign Migration

In y-pol tight px60, Δp_x sign migration is small compared with the neutron visibility effect.

Backup C: Response Matrices



Response matrices are useful for closure and possible unfolding, but the main result does not rely on matrix inversion.

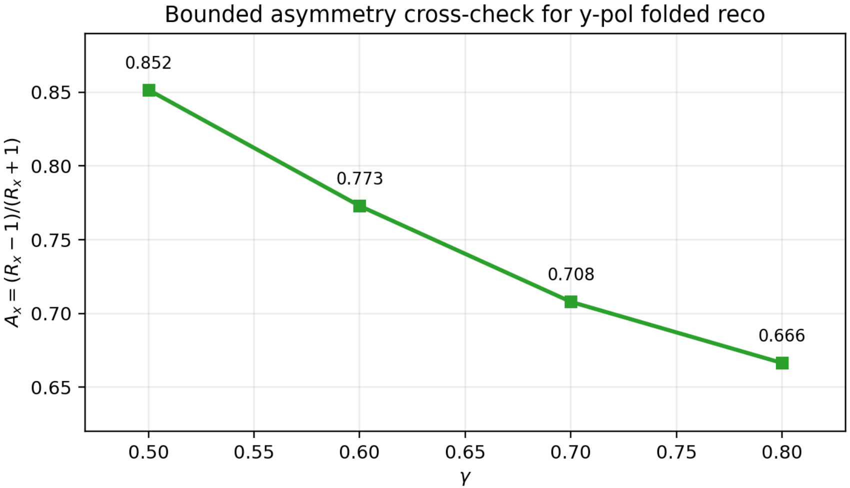
Backup D: ^{124}Sn Full y-pol Numbers

γ	N_+	N_-	$R_x^{\text{reco,fid}} \pm \sigma_{\text{MC}}$	A_x	N_{used}
0.5	2134	171	12.48 ± 0.99	0.852	2305
0.6	2147	275	7.81 ± 0.50	0.773	2422
0.7	2093	358	5.85 ± 0.33	0.708	2451
0.8	1948	390	4.99 ± 0.28	0.666	2338

Meaning

These are folded reco-plane values for ^{124}Sn y-pol tight px60, not truth-level asymmetries. Uncertainties are current MC statistics only.

Backup D: Bounded Asymmetry Cross-Check



A_x is bounded and monotonic for the same reconstructed y-pol points. It is a stability cross-check, not a replacement for the primary R_x result.

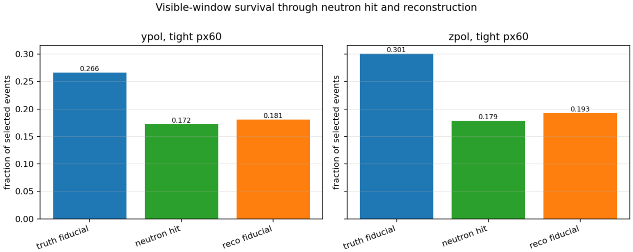
Backup D: Adjacent Gamma Separation

interval	R_{left}	R_{right}	$\lvert\Delta R\rvert$	$N_{\mathrm{stat},3\sigma}$
0.5→0.6	12.48	7.81	4.67	519
0.6→0.7	7.81	5.85	1.96	979
0.7→0.8	5.85	4.99	0.85	2.77×10^3

Scope

This is a statistical-only planning scale, not an experimental significance claim.

Backup D: Usable-Event Survival



Planning formula:

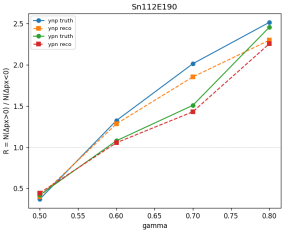
$$N_{\mathrm{usable}} = \mathrm{rate} \times T \times \frac{N_{\mathrm{reco\ plane}}}{N_{\mathrm{raw}}}.$$

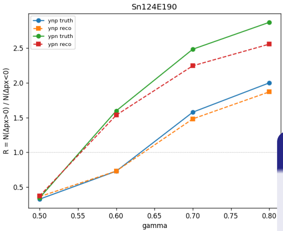
For y-pol:

$$\frac{N_{\mathrm{reco\ plane}}}{N_{\mathrm{raw}}} = 0.00566$$

$$N_{\mathrm{usable}} \simeq 2.75 \times 10^5.$$

Backup E: ^{112}Sn Physics-Motivation Reference





Existing Sn112 information is useful as a physics-motivation reference, but not as a direct quantitative comparison with the current Sn124 NEBULA Plus closure.

Reason

The locally exported tight-px60 folded tables are Sn124-only; older Sn112 ROOT files come from a different reconstruction production.

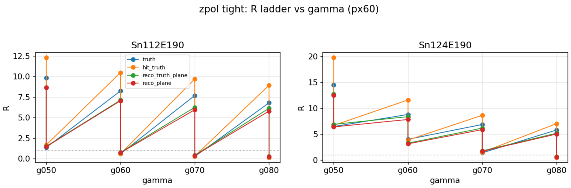
Backup E: Sn112/Sn124 Consistency Audit

item	current Sn124 result	Sn112 direct comparison
production	NEBULA Plus joint 20260609	described, not exported locally
geometry	SAMURAI + NEBULA Plus	intended same
backend	PDC NN + neutron time-of-flight	intended same
y-pol γ grid	0.5,0.6,0.7,0.8	intended same
cut/fiducial	truth-assisted tight + reco px60	not verified locally
event plane	reconstructed	not verified locally

Decision

Do not show a direct quantitative isotope plot until Sn112 is exported and checked with the identical chain.

Backup F: z-pol Folded Results



γ	N_+	N_-	R_{reco}
0.5	306	48	6.38
0.6	208	67	3.10
0.7	214	129	1.66
0.8	171	307	0.56

z-pol remains physically valuable but is kept as a second-stage extension.

Backup F: y-pol and z-pol Cut Definitions

y-pol tight

$$|p_{y,p}^{\mathrm{truth}} - p_{y,n}^{\mathrm{truth}}| < 150$$

$$50 < |\vec{p}_{T,p}^{\mathrm{truth}} + \vec{p}_{T,n}^{\mathrm{truth}}| < 200$$

$$\pi - |\phi_{\mathrm{truth}}| < 0.2.$$

z-pol tight

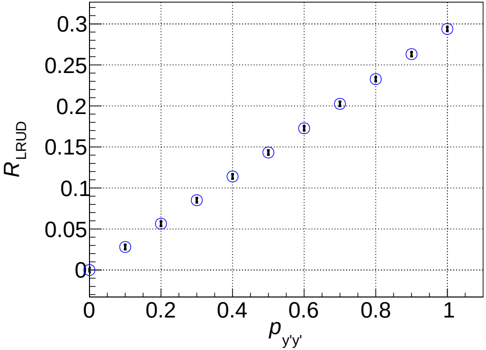
$$p_{z,p}^{\mathrm{truth}} + p_{z,n}^{\mathrm{truth}} > 1150$$

$$|p_{z,p}^{\mathrm{truth}} - p_{z,n}^{\mathrm{truth}}| < 150$$

$$|\vec{p}_{T}^{\mathrm{sum}}| > 50, \quad \pi - |\phi_{\mathrm{truth}}| < 0.5.$$

These current event-quality cuts are truth-assisted and must be migrated to reco-only cuts.

Backup G: y-pol Polarization Monitoring

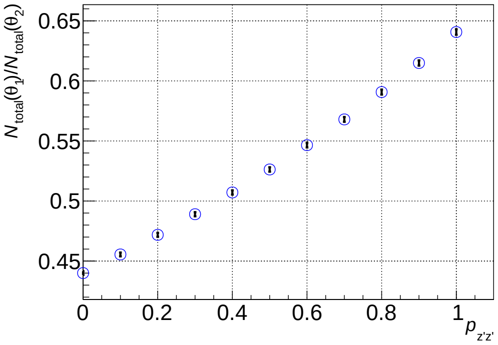


Vertical tensor polarization p_{yy} can be monitored through an established LRUD asymmetry in deuteron-proton elastic scattering.

Role in IVR

Polarization monitoring is an independent prerequisite and must be propagated into $R_x(\gamma)$.

Backup G: z-pol Monitoring Concept



Longitudinal tensor polarization p_{zz} can be monitored with a more demanding absolute/two-angle normalization strategy.

Careful statement

p_{zz} monitoring is not impossible; it is more complex than the y-pol LRUD baseline.

Backup H: Truth-Assisted Cuts

What is fully reconstructed now

Numerator and denominator of $R_x^{\mathrm{reco,fid}}$ use reconstructed proton momentum, reconstructed neutron momentum, and reconstructed event plane.

What still uses truth

Event-quality cuts such as transverse balance and near-backscatter selection are currently defined using truth momenta in simulation.

Required closure

Re-run with reco-only cuts and compare efficiency, purity, migration, and γ ordering against the truth-assisted baseline.

Backup H: Detector-Systematic Checklist

- neutron threshold, T_0, timing resolution, and hit-position model,
- cross-talk, fake neutron candidates, and accidental coincidences,
- charge-veto and charged-particle leakage assumptions,
- detector geometry and alignment variations,
- background and pile-up in the relevant electronics time window,
- polarization uncertainty and model/folding uncertainty.

Next analysis product

Pseudo-data closure under detector and model variations, with the same folded comparison used for the experimental observable.